

1.3.1

a. $A[0..5] = [60, 35, 81, 98, 14, 47]$

$i=0, \text{Count}[0..5] = [3, 0, 1, 1, 0, 0]$

$i=1, \text{Count}[0..5] = [3, 1, 2, 2, 0, 1]$

$i=2, \text{Count}[0..5] = [3, 1, 4, 3, 0, 1]$

$i=3, \text{Count}[0..5] = [3, 1, 4, 5, 0, 1]$

$i=4, \text{Count}[0..5] = [3, 1, 4, 5, 0, 2]$

$S[\text{Count}[i]] \leftarrow A[i], S = [14, 35, 47, 60, 81, 98]$

b. 不稳定 若 $A[i] = A[j], i < j$

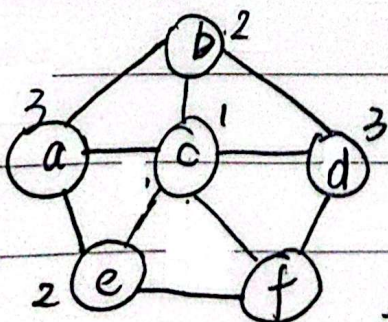
则 $\text{Count}[i] > \text{Count}[j]$

则 S 中 i 在 j 后, 位置相比于输入互换了.

故不稳定.

c. 不在位, 引入了两个额外的数组 Count 和 S .

1.3.8 (a). 该图对应的顶点. 边:



(b) 令 $c=1, b=2$

则 $a=3, d=3, e$ 取 2

此时 f 只能取 $f=4$.

故一共最少用 4 种颜色着色

2.1.10

a. $\sum_{i=1}^{64} 2^{i-1} = 2^0 + 2^1 + 2^2 + \dots + 2^{63} = 2^{64} - 1$, 需要 $(2^{64} - 1)$ 秒.

b. $\sum_{i=1}^{64} (2i-1) = 1 + 3 + 5 + 7 + \dots + 2 \times 64 - 1 = 64^2 = 2^{12} = 4096$, 需要 4096 秒.

2.2.5 对 $(n-2)!$, $5 \lg(n+100)^{10}$, 2^{2n} , $0.001n^4 + 3n^3 + 1$, $\ln^2 n$, $\sqrt[3]{n}$, 3^n 排序

$(n-2)! \in \Theta((n-2)!)$, $5 \lg(n+100)^{10} = 50 \lg(n+100) \in \Theta(\lg n)$

$2^{2n} \in \Theta(4^n)$, $0.001n^4 + 3n^3 + 1 \in \Theta(n^4)$, $\ln^2 n \in \Theta(\lg^2 n)$

$\sqrt[3]{n} \in \Theta(n^{\frac{1}{3}})$, $3^n \in \Theta(3^n)$

$\Theta(\lg n) < \Theta(\lg^2 n) < \Theta(n^4) < \Theta(4^n) < \Theta((n-2)!)$

$\lim_{n \rightarrow \infty} \frac{n^4}{3^n} = \lim_{n \rightarrow \infty} \frac{4n}{n \cdot 3^{n-1}} = 0$, 故 $\Theta(n^4) < \Theta(3^n)$

$\lim_{n \rightarrow \infty} \frac{\lg^2 n}{n^{\frac{1}{3}}} = \lim_{n \rightarrow \infty} \frac{2 \lg n \cdot \frac{1}{\ln 10 \cdot n}}{\frac{1}{3} \cdot n^{-\frac{2}{3}}} = \lim_{n \rightarrow \infty} \frac{2 \lg n}{\frac{1}{3} \cdot \ln 10 \cdot n^{\frac{1}{3}}} = \lim_{n \rightarrow \infty} \frac{2 \cdot \frac{1}{\ln 10} \cdot \frac{1}{n}}{\frac{1}{3} \cdot \ln 10 \cdot n^{\frac{2}{3}}} = 0$

故 $\Theta(\lg n) < \Theta(\lg^2 n) < \Theta(n^{\frac{1}{3}}) < \Theta(n^4) < \Theta(3^n) < \Theta(4^n) < \Theta((n-2)!)$

由低到高排序: $5 \lg(n+100)^{10}$, $\ln^2 n$, $\sqrt[3]{n}$, $0.001n^4 + 3n^3 + 1$, 3^n , 2^{2n} , $(n-2)!$